

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – CONCEPT MAPPING

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	Assessment:	Assessment Tool 2 – Concept Mapping
Weightage:	20% of CIA	Total Marks:	100 (2 Questions × 50 Marks)
CO1 Statement:	<i>Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)</i>		
Student Name:	P. Sai Charan Tej	Reg. No.:	192465048
Section / Batch:		Date:	

Instructions to Students

- Answer BOTH questions. Each question carries 50 Marks. Total: 100 Marks.
- Draw a clear and neat concept map for each question.
- Identify all key concepts and arrange them in a logical hierarchy.
- Show meaningful linkages between concepts using arrows and connecting words.
- Compare related concepts wherever required and justify the relationships clearly.
- Ensure the concept map is well-organized, readable, and properly labelled.

Question Type	Focus Area	Questions	Marks
Concept Mapping	Map algorithm efficiency concepts using asymptotic analysis, search techniques, recurrence relations, and recursion tree method	Q1 – Q2	Q1 –50 Q2 –50
GRAND TOTAL:		100 Marks	

SECTION A – CONCEPT MAPPING

Q43. Consider the concept map: Algorithm → Space Complexity → Auxiliary Space → Input Space Map how memory usage is calculated. Extend the map with time-space trade-off. Justify when higher space is acceptable. Interpret how the map helps design efficient solutions.

Q44. Consider the concept map: Recursive Algorithm → Recursion Depth → Call Stack → Space Complexity Explain how recursion affects memory usage. Extend the map with input size influence. Justify why deep recursion increases space. Interpret how this helps optimize memory.

Code of Conduct

I P. Sai Charan Tej(192465048)__(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

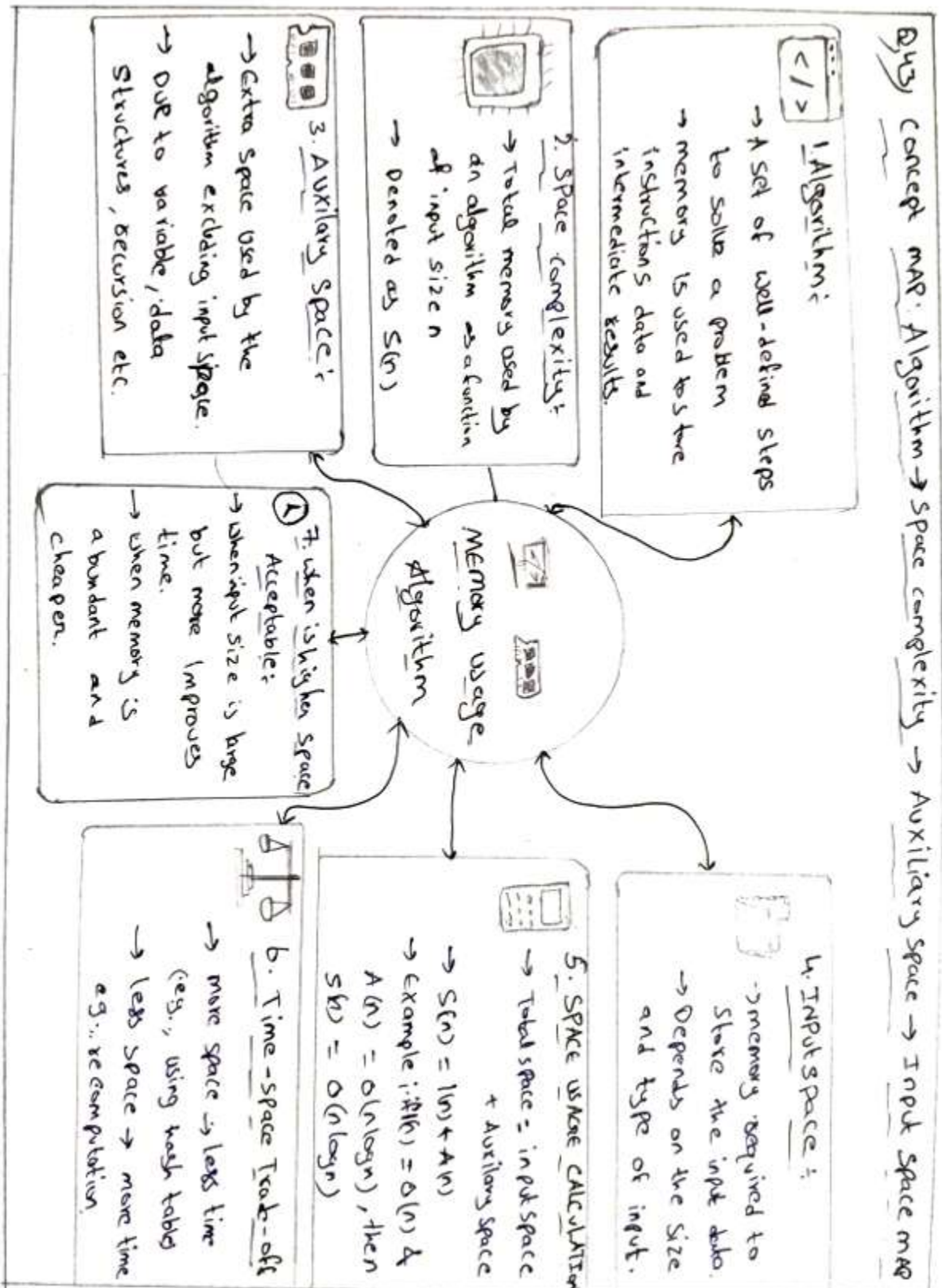
Marks Evaluation:

Question No.	Concept Identification (25%)	Linkages (25%)	Organization (20%)	Integration of Literature (20%)	Clarity (10%)	Total Marks (Each – 50)
Q1						
Q2						
Overall Total (100)						

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

S.N	Register 192465048	Student Name P. Sai Charan Tej	Assigned Q43	Assigned Q44
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CONCEPT MAPING



4th) Recursive Algorithm → Recursion Depth → Call Stack → Space complexity → Concept map



1. Recursive Algorithm:

- A function calls itself to solve smaller instance of the same problem.
- Stops when a base case is reached.

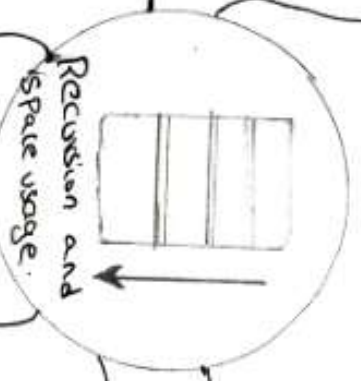


2. Recursion Depth:

- Maximum number of nested recursive calls
- Depends on input size n .

5. input size influence

Type of Recurrence	Recursion Depth	Space complexity
$T(n) = T(n-1) + O(1)$	n	$O(n)$
$T(n) = T(n/2) + O(1)$	$\log n$	$O(\log n)$
$T(n) = T(n/5) + O(1)$	n/c	$O(n)$
Balance with $\log n$	$\log n$	$O(\log n)$



3. Call Stack:

- Each recursive call is pushed onto the stack
- Stores: Parameters, return address, local variables
- Popped when the call returns



4. Space Complexity:

- Space used: call stack space + any auxiliary space
- $S(n) = O(\text{depth}) + O(\text{auxiliary space})$
- For most recursive algorithms $S(n) = O(\text{recursion depth})$



b. Why deep recursion increase space?

- Each call creates a new stack frame
- more depth → more frames on stack
- more frames → more memory used
- If depth is too long → stack overflow

